Multi-asset benchmark — initial results

Coinbase BTC / ETH / SOL, 7 days Jan 2026 — final_v2 protocol per coin

Snapshot 2026-07-13. 6 models × 3 training seeds × 3 rollout seeds per coin; seq/stride 1024, 12 epochs (equal step counts across models within each coin), val-split rate calibration with full-scale verification (SAHP uncalibrated by protocol, †), lossless streaming prediction, carried-state rollouts, equal-duration real-vs-sim comparison. **This is a partial snapshot**: reruns/retries in flight (see §4).

1. Run status

coin	complete	train fail (infra)	SF fail (calibration)	running (incl. reruns)
btc	14	0	3	1
eth	3	0	1	14
sol	15	0	3	0

Train failures are infrastructure only (NFS stale file handle during the first-launch tensor-cache race, twice "CUDA device busy"); reruns (already submitted, in flight at snapshot time) re-read the now-existing caches. SF failures are *substantive* calibration failures, all in the S2P2 family (§4).

2. Prediction (per genuine event; mean ± 95% CI over available seeds)

Bold = best per column: lower is better except ACC (higher) and mean_u (closest to 1).

BTC

model	overall	timeNLL	markNLL	KS	mean_u	tMAE	ACC	PPL
nhp	**-1.067±0.014**	**-3.215±0.004**	**2.148±0.016**	**0.227±0.006**	1.07±0.00	**0.03±0.00**	**0.449±0.003**	**8.57±0.13**
lstm	-0.437±0.400	-2.710±0.343	2.273±0.063	0.305±0.034	1.44±0.54	0.03±0.00	0.411±0.015	9.71±0.62
sahp (n=2)	-0.928±0.071	-3.188±0.093	2.260±0.022	0.258±0.116	0.89±0.20	0.03±0.01	0.412±0.006	9.58±0.21
pct-lstm	-0.598±0.079	-3.176±0.048	2.579±0.035	0.242±0.042	**1.01±0.11**	0.03±0.00	0.313±0.010	13.18±0.47
s2p2	-0.397±0.235	-2.578±0.245	2.181±0.019	0.330±0.037	2.36±0.21	1.47±2.69	0.439±0.003	8.86±0.17
ss2p2-full	-0.865±0.109	-3.018±0.116	2.152±0.008	0.265±0.001	1.85±0.13	0.03±0.00	0.447±0.003	8.61±0.07

ETH

model	overall	timeNLL	markNLL	KS	mean_u	tMAE	ACC	PPL
nhp (n=1)	**-0.813**	**-3.413**	**2.600**	**0.260**	**1.01**	**0.03**	**0.342**	**13.46**
lstm (n=1)	0.138	-2.618	2.756	0.334	1.66	0.03	0.288	15.74
sahp (n=1)	-0.596	-3.317	2.721	0.302	0.85	0.04	0.299	15.20
pct-lstm	–	–	–	–	–	–	–	–
s2p2 (n=1)	-0.433	-3.055	2.623	0.308	2.20	3.68	0.333	13.77
ss2p2-full	–	–	–	–	–	–	–	–

SOL

model	overall	timeNLL	markNLL	KS	mean_u	tMAE	ACC	PPL
nhp	0.528±0.014	-2.707±0.004	3.235±0.014	0.321±0.008	**1.15±0.06**	0.05±0.00	0.184±0.004	25.40±0.35
lstm	1.343±0.250	-2.067±0.218	3.410±0.040	0.420±0.025	1.56±0.45	0.06±0.00	0.141±0.005	30.28±1.22
sahp	0.668±0.024	-2.687±0.027	3.355±0.019	0.376±0.017	0.83±0.04	0.06±0.00	0.155±0.003	28.65±0.54
pct-lstm	0.895±0.028	-2.607±0.026	3.502±0.009	0.314±0.010	1.23±0.02	0.06±0.00	0.105±0.004	33.18±0.30
s2p2	0.671±0.494	-2.560±0.495	3.231±0.001	0.204±0.017	2.08±0.52	46.18±95.99	0.182±0.002	25.30±0.03
ss2p2-full	**0.274±0.047**	**-2.911±0.044**	**3.185±0.003**	**0.173±0.007**	1.57±0.05	**0.05±0.00**	**0.192±0.001**	**24.17±0.07**

3. Stylized facts (rel-err vs real; rollout seeds averaged per checkpoint; mean ± 95% CI over checkpoints)

Checkpoints whose calibration failed verification produce no SF files and are excluded with a note; † = SAHP reported uncalibrated (k=1) by protocol.

BTC

model	sim_rate	rate_re	Fano_re	clus_re	retACF_re	cal_k
nhp	37.92±2.81	0.086±0.080	0.863±0.069	**0.275±0.042**	**0.209±0.138**	0.800±0.090
lstm	38.51±3.24	0.103±0.093	0.993±0.000	0.495±0.404	0.537±0.447	0.472±0.060
sahp † (n=2)	7.67±35.93	0.780±1.029	0.791±0.810	0.334±1.687	0.245±1.922	1.000±0.000
pct-lstm	37.13±1.90	**0.063±0.054**	**0.786±0.119**	0.360±0.288	0.333±0.321	0.728±0.092
s2p2 (n=1)	33.74	0.069	1.114	0.326	0.642	0.541
ss2p2-full (n=2)	43.29±63.90	0.240±1.830	0.788±0.992	0.462±0.618	0.384±1.139	0.608±0.167

Excluded (calibration failed at matched fidelity): s2p2-s1, s2p2-s2, ss2p2-full-s3

ETH

model	sim_rate	rate_re	Fano_re	clus_re	retACF_re	cal_k
nhp (n=1)	45.65	**0.307**	**0.841**	0.599	**0.188**	0.805
lstm (n=1)	48.55	0.389	0.994	0.876	0.732	0.545
sahp † (n=1)	9.22	0.736	0.871	**0.366**	0.498	1.000
pct-lstm			*no calibrated checkpoints*			
s2p2			*no calibrated checkpoints*			
ss2p2-full			*no calibrated checkpoints*			

Excluded (calibration failed at matched fidelity): s2p2-s1

SOL

model	sim_rate	rate_re	Fano_re	clus_re	retACF_re	cal_k
nhp	23.93±0.69	**0.192±0.034**	0.751±0.167	0.721±0.109	0.806±0.071	0.799±0.025
lstm	24.66±0.75	0.228±0.037	0.988±0.000	0.969±0.032	0.861±0.100	0.500±0.000
sahp †	6.40±3.02	0.681±0.150	0.925±0.078	**0.656±0.241**	0.859±0.063	1.000±0.000
pct-lstm	24.58±4.80	0.224±0.239	**0.460±0.335**	0.715±1.356	1.665±2.536	0.626±0.180
s2p2			*no calibrated checkpoints*			
ss2p2-full	26.08±3.37	0.298±0.168	0.598±0.599	0.683±0.624	**0.778±0.156**	0.591±0.080

Excluded (calibration failed at matched fidelity): s2p2-s1, s2p2-s2, s2p2-s3

4. Calibration findings and pending work

s2p2 near-criticality replicates across assets. Six of nine s2p2 checkpoints failed val-side calibration (btc s1,s2; eth s1; sol s1,s2,s3), with the same signature as the Gemini ETH benchmark: a near-discontinuous closed-loop rate response in the time-rescaling constant k (e.g. btc-s1: k 0.5930→0.5938 flips the probe rate 36→48.5 ev/s) and full-scale verify misses of 30–143%. On SOL *all three* s2p2 seeds failed — pending one RESUME retry, s2p2 may have no calibrated simulation arm on that asset (a model-level statement, as with SAHP). ss2p2-full calibrated and verified everywhere except btc-s3 (probe bisection non-convergence near a steep spot; retry pending). **In flight at snapshot**: 14 infrastructure train reruns (13 eth, 1 btc) and 7 RESUME SF retries, already submitted; 1 first-round task still training; strict per-coin reports (`REPORT_OK`) run after those land.

Reading the snapshot. Coinbase rates are roughly 6–13× Gemini ETH (val rates: btc 38.3, eth 47.9, sol 24.0 ev/s), so the fixed 1024-event context spans only ∼20–40s of market time here vs ∼4.5 min on Gemini — long-memory facts (clus/retACF) lean harder on carried state and TBPTT. Numbers are comparable *within* a coin; cross-coin comparisons should use relative errors only. CIs over fewer than 3 checkpoints are marked with their n; n=2 uses $t_{0.975}$=12.7 and is wide by construction.